BYTEATHON 2026

GRAND FINALE

Problem Statements Handbook

Organized by Department of Computer Science and Engineering,
Noorul Islam Centre for Higher Education

DURATION & DATE
24-Hour | 15–16 Oct 2026

TRACK SELECTION
Top 5 Curated Tracks

TRACK OVERVIEW

TRACK ID	DOMAIN	OFFICIAL TITLE
BYTE01	Agriculture & Food Supply	Intelligent Post-Harvest Loss Prevention
BYTE02	Law, Governance & Public Services	Explainable Policy Interpretation
BYTE03	Accessibility & Inclusive Tech	Automated Accessibility Barrier Remediation
BYTE04	Database & Data Engineering	Self-Adapting Data Pipelines
BYTE05	AI & Machine Learning	Detecting Silent Reliability Failures

BYTE01

AGRICULTURE & FOOD SUPPLY CHAIN

CODENAME: HARVEST ZERO

Intelligent Post-Harvest Loss Prevention and Distribution Optimization

- **PROBLEM STATEMENT**

> Every year, agricultural produce is lost between harvest and consumption because of inefficient distribution, unpredictable demand, limited storage facilities, and logistical delays.

- **THE CHALLENGE**

Develop an intelligent decision-support platform that connects farmers, storage facilities, transport operators, and buyers to minimize post-harvest losses. The system must consider produce perishability, transportation constraints, fluctuating demand, storage capacity, and economic viability to recommend optimal distribution strategies.

OPTIONAL N8N INTEGRATION
Automated inventory updates, buyer notifications, logistics coordination, and spoilage-risk alerts.

BYTE02

LAW, GOVERNANCE & PUBLIC SERVICES

CODENAME: THE POLICY MAZE

Explainable Policy Interpretation and Eligibility Decision Support

PROBLEM STATEMENT

> Government schemes, institutional regulations, and public benefit programs frequently contain overlapping eligibility requirements, exceptions, conflicting documents, and amendments. As a result, determining whether an individual qualifies for a particular benefit can be complicated, time-consuming, and prone to error.

THE CHALLENGE

Develop an intelligent platform capable of interpreting policy documents, extracting requirements, resolving applicable exceptions, and evaluating eligibility based on supporting information. The platform must provide traceable explanations for every decision, handle incomplete documentation, and recognize policy changes without relying on unsupported assumptions.

OPTIONAL N8N INTEGRATION

Document ingestion, approval routing, amendment notifications, and application status workflows.

BYTE03

ACCESSIBILITY & INCLUSIVE TECHNOLOGY

CODENAME: THE INVISIBLE BARRIER

Automated Detection and Remediation of Accessibility Barriers in Digital Services

PROBLEM STATEMENT

> Millions of users encounter barriers when accessing digital platforms because of inaccessible navigation, poorly structured interfaces, insufficient visual contrast, and incompatible interaction patterns. Although automated accessibility testing exists, identifying issues does not necessarily translate into meaningful improvements in real user workflows.

THE CHALLENGE

Build an intelligent accessibility remediation platform that identifies barriers within web applications, prioritizes their practical impact, and recommends or implements verifiable corrections. Solutions must demonstrate improvements in functional usability rather than relying solely on automated compliance scores.

OPTIONAL N8N INTEGRATION

Scheduled accessibility audits, issue reporting, remediation ticket creation, and regression alerts.

BYTE04

DATABASE SYSTEMS & DATA ENGINEERING

CODENAME: SCHEMA SHIFT

Building Self-Adapting Data Pipelines for Inconsistent and Evolving Data Sources

PROBLEM STATEMENT

> Organizations depend on data collected from multiple databases, applications, APIs, and external services. However, these sources frequently differ in structure, formatting, naming conventions, and reliability. Unexpected schema changes, missing fields, duplicate records, and inconsistent updates can silently corrupt downstream analytics and business processes.

THE CHALLENGE

Develop an adaptive data integration platform capable of ingesting heterogeneous datasets, detecting structural changes, identifying inconsistencies, and maintaining a reliable unified data representation. The platform should provide traceability for transformations, preserve original data, and allow recovery from incorrect mappings or failed processing operations.

OPTIONAL N8N INTEGRATION

Source ingestion workflows, validation triggers, failed-record routing, data-quality alerts, and synchronization.

BYTE05

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING ENGINEERING

CODENAME: CONFIDENCE COLLAPSE

Detecting and Responding to Silent Reliability Failures in Deployed Machine Learning Systems

PROBLEM STATEMENT

> Machine learning models are typically evaluated using historical datasets before deployment. However, real-world data distributions, user behavior, and operating environments change continuously. As conditions evolve, a previously reliable model may produce increasingly inaccurate predictions while continuing to report high confidence.

THE CHALLENGE

Develop an intelligent model reliability monitoring system capable of detecting distribution shifts, identifying suspicious prediction behavior, estimating reliability degradation, and recommending corrective actions. The system must distinguish meaningful changes from normal fluctuations and support safe responses such as confidence-based abstention, model fallback, or retraining recommendations.

OPTIONAL N8N INTEGRATION

Automated drift alerts, model review workflows, approval-based retraining, and monitoring reports.